

MAR DEL PLATA, Argentina

WMO: 876920

Lat: 37.9322S

Lon: 57.5814W

Elev: 21

StdP: 101.07

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.1	0.1	-4.7	2.5	3.5	-3.0	2.9	3.6	11.6	10.2	10.4	9.8	2.0	320	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	31.3	21.2	29.3	20.5	27.5	20.0	23.2	28.0	22.3	26.5	21.5	25.3	6.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	16.6	24.9	21.0	15.7	24.0	20.2	14.9	23.2	69.1	27.8	65.6	26.4	62.7	25.4	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.5	8.5	-3.4	35.7	1.4	1.7	-4.3	36.9	-5.1	37.8	-5.9	38.8	-6.9	40.0
			-3.9	25.3	1.4	1.4	-4.9	26.3	-5.8	27.1	-6.5	27.9	-7.6	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	20.3	19.7	18.0	14.6	11.7	8.7	7.9	9.3	10.4	13.2	15.8	18.8
	DBStd	5.32	3.27	3.27	3.32	3.13	3.07	2.76	2.90	3.08	2.96	3.19	3.31	3.66
	HDD10.0	242	0	0	0	2	17	57	78	51	30	8	1	0
	HDD18.3	1833	18	22	49	117	207	288	322	281	238	163	89	39
	CDD10.0	1704	319	271	247	142	69	19	14	28	42	106	176	273
	CDD18.3	254	79	59	37	6	1	0	0	1	0	3	14	54
	CDH23.3	1869	572	378	204	37	6	0	0	6	4	39	143	481
	CDH26.7	577	191	115	48	4	0	0	0	1	1	5	33	179
(14) Wind	WSAvg	4.0	4.5	4.2	3.9	3.6	3.3	3.7	3.7	4.0	4.2	4.4	4.5	4.7
(15) Precipitation	PrecAvg	902	96	78	99	85	65	60	56	65	58	84	77	95
	PrecMax	1405	206	303	267	370	239	174	173	195	157	254	188	331
	PrecMin	558	21	11	8	1	5	2	0	2	5	6	18	6
	PrecStd	191	50	49	55	80	43	41	38	51	36	50	40	54
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	32.9	30.9	26.8	23.9	19.0	19.2	23.1	23.0	27.1	30.1	34.0
		MCWB	22.3	22.5	21.4	19.4	17.8	14.6	14.6	16.4	15.8	18.5	19.2	21.9
	2%	DB	31.0	30.0	27.9	24.1	20.4	16.9	16.4	19.0	20.1	24.0	27.1	31.0
		MCWB	21.6	21.4	20.7	18.2	16.0	13.2	12.8	13.9	13.9	16.8	18.6	20.5
	5%	DB	28.7	27.7	25.5	22.1	18.5	15.2	14.5	16.7	18.0	21.5	24.7	28.3
		MCWB	20.7	20.7	19.7	17.2	14.7	12.2	11.3	12.0	12.9	15.6	17.6	19.7
	10%	DB	26.4	25.3	23.6	20.2	17.0	14.0	12.9	14.8	16.0	19.1	22.2	25.9
		MCWB	20.0	20.3	19.3	16.9	14.2	11.4	10.5	11.0	11.9	14.4	16.7	19.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	24.4	23.4	21.4	18.9	17.1	16.6	17.9	17.4	20.3	21.1	23.7
		MCDB	30.0	29.1	27.1	23.8	21.8	17.9	18.4	20.9	20.7	25.0	26.4	30.4
	2%	WB	23.0	23.1	22.1	19.9	17.2	14.8	13.8	14.9	15.2	17.9	19.7	22.2
		MCDB	28.1	27.1	25.5	21.8	19.2	15.8	15.3	17.5	18.4	22.4	24.6	27.9
	5%	WB	22.1	22.1	21.0	18.6	16.0	13.2	12.1	13.1	13.9	16.6	18.6	21.0
		MCDB	26.5	25.5	23.9	20.8	17.5	14.4	13.5	15.4	16.7	20.1	22.8	25.8
	10%	WB	21.2	21.2	20.1	17.4	14.8	11.8	11.0	11.7	12.7	15.2	17.6	19.8
		MCDB	25.2	24.3	22.7	19.8	16.5	13.5	12.5	13.9	15.0	18.2	21.3	24.1
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	10.1	9.9	10.1	9.5	9.3	9.1	9.5	9.7	9.6	10.7	11.4
		MCDBR	15.2	14.0	13.0	13.1	11.9	11.0	11.4	13.2	13.5	14.6	15.2	16.5
	5% WB	MCWBR	7.4	7.0	6.9	7.7	7.7	7.7	7.8	8.3	8.3	8.4	8.1	7.8
		MCWBR	12.2	10.9	10.1	10.1	9.2	8.1	8.6	10.4	11.2	11.9	12.3	13.3
(40) Clear-Sky Solar Irradiance	taub		0.402	0.384	0.363	0.356	0.346	0.339	0.347	0.397	0.419	0.391	0.376	0.386
		taud	2.362	2.429	2.481	2.451	2.445	2.451	2.402	2.226	2.176	2.321	2.394	2.384
	Ebn at Noon		930	924	903	842	785	758	772	777	827	905	950	951
		Edh at Noon	130	116	102	92	80	74	82	113	135	128	125	128
(44) All-Sky Solar Radiation	RadAvg		7.21	6.23	4.91	3.53	2.39	1.92	2.07	2.82	4.08	5.36	6.64	7.53
	RadStd		0.30	0.42	0.31	0.36	0.21	0.17	0.18	0.30	0.31	0.52	0.34	0.38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	+29
(47) Regional (2 neighbors)		N/A	N/A	N/A	+0.70	N/A	N/A	N/A	N/A	+33

Nomenclature: See separate page